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NEEDHAM GRAVITY

PROPOSAL - MS/HS RC

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Needham Gravity, the Needham NASA HERC team, consists of seven sophomore friends from Needham High School. In the past, three of us on the team have participated and came in first place in the MIT BeaverWorks' Build a CubeSat National Challenge. All of us are passionate about STEM, robotics, and engineering. We are excited to have the chance to work together as a team to accomplish something we would all be immensely proud of. We are all highly motivated and committed to building a well-designed rover, the plans for which we will detail below.

1. Facilities And Equipment

1.1 Facilities: To construct the rover and its custom components (such as the chassis, dual-suspended track system, and any others that we may require as dictated by the NASA HERC course and task), we will use the DaVinci Workshop (DVW), which is the makerspace within Needham High School. It has plenty of floor space, equivalent to about three classrooms, for discussion, construction, and testing. We will have free access through our Faculty Advisor, Hans Batra, the current DVW Director. If this makerspace is not sufficient, we can also access makerspaces at Olin College of Engineering. Hans Batra has connections at Olin, so he can secure makerspace access for us. We will likely access the Olin makerspaces through an in-kind donation, and we likely will not have to pay.

Equipment: In the DaVinci Workshop, the team will have access to: five 3D printers (Bambu Carbon X1C, Prusa i3 MK3S+, Prusa MK4S, Ultimaker 3 and 3E) and filament; a Trotec 30 Watt Speedy 300 laser cutting and engraving machine; an isolated testing field (12ft x 12ft); various prototyping materials like foam boards or popsicle sticks; basic hand tools; basic power tools; and a plethora of components, some from previous Robotics years. In Olin's two makerspaces (The Bot Shop and The Machine Shop), we will have access to: a Hurco VM10i 3-axis CNC machine; six Bridgeport-style hybrid manual/CNC mills; five manual lathes; an OMAX waterjet cutter; three welding booths with MIG, TIG, Stick, and Oxy-Acetylene setups; 48-inch hydraulic shear and brake for metal sheet manipulation; Markforged Onyx Series printers; and various other 3D printers, laser CNC machines, and other tools.

Accessibility: The DVW will be accessible to our team after school 2-3 days a week. All team members have experience with its tools and materials from past projects. Our team can also request additional time in the DaVinci Workshop. We will have full access to these materials and machines during the allotted times, except for powered tools, which require explicit permission from a supervising mentor or teacher. Some tools will likely require minimal training (e.g., drill press, laser CNC/engraving machine, etc.) beyond what is already provided for any use of the DVW. If needed, a mentor (Mr. Batra or another volunteer) or another senior member of Needham Robotics can provide training. For Olin, we will likely have access to these makerspaces after school (2:35 PM) on a first-come, first-served basis. Training there will be provided by university students using the makerspaces with us.

1.2 As a subset of the long-running Needham Robotics Club, Needham Gravity can leverage existing connections with various companies for mentorship or funding. At Needham High School, besides our advisor, Mr. Batra, other teachers, such as physics teachers Mr. Smalley and Mr. Guerra, are actively involved in robotics. Mr. Batra is also a Technician-class licensed ham radio operator, which will let us test our task execution by transmitting AX.25 radio data. These teachers can guide our team when issues come up and are specifically instructed to structure their help so we can find the answers ourselves. By doing so, they help the information stick better and help us learn to problem-solve instead of being told what to do.

2. Safety

2.1.1 To maintain safety throughout the NASA HERC program, the team shall perform hazard analysis using Failure Modes and Effects Analysis (FMEA) before working on the rover. The Safety Officer shall share a spreadsheet listing each material type, and the team shall identify how each tool, possible component, and other material may fail in the second column. Then, under the Safety Officer's direction, the team shall identify all possible consequences of the failure and list them in the third column. After completing this analysis, the team shall rank the material by hazard likelihood and average severity using a one-to-ten scale (least to most). Multiply these two scales to get the RPN (Risk Priority Number), which will determine the definitive ranking of the materials by hazard level. Each hazard shall also be categorized by the effect/hazard posed if it fails or is misused; for example, whether it can cause a cut or is toxic (airborne or drinkable). Below is an example line of this type of spreadsheet:

Tool / Material	Possible Failures	Hazards	Avg. Likelihood	Avg. Severity	RPN	Safety Equipment
Track Saw	May not be connected to dust port, overheating, clamps misplaced or slip, kickback from blade binding, overextension of blade	Dust may be inhaled, dirty workspace, saw can short-circuit and cause sparks leading to fire if wood is nearby, saw or material can hit the operator violently	4	5	20	Safety goggles in case of debris, knowledge of the location of fire extinguishers

Accordingly, the team will draft possible contingency plans, including what to do if someone gets hurt, and identify the safety equipment needed to mitigate the risks of both failure effects and material failure. Additionally, in an emergency, the team shall exit through the front door of the lab under the Safety Officer's supervision and then make way for the door to the right of the classroom, which leads outside, and escape by these means.

2.1.2 As mentioned in the Facilities and Equipment section, we shall use the DaVinci Workshop at Needham High School to design, construct, and test the rover. As described above, the DaVinci Workshop has five 3D printers, a laser CNC and engraving machine, and various other power tools. We will therefore also use the DaVinci Workshop Safety Plans and documentation as guidelines for all DaVinci Workshop materials. One of the most general rules in the DaVinci Workshop is that an adult must always be in the room, and we will abide by it. Additionally, for the use of the previously referenced power tools and hand tools, such as the laser CNC or 3D printers, the user must be trained before using them (some team members already have training through the Robotics Club at the school, which utilizes the same space) and must get verbal consent before using them. During any lab work, members will wear safety glasses and follow this rule strictly when using power tools. For all industrial equipment, the two-person rule applies: two people must work together to use a tool. Moreover, if any equipment seems to be malfunctioning or having issues, the student shall stop using it and inform the advisor so it can be fixed and no one gets hurt.

For safety, the team shall follow the tool-specific safety rules in this [document](#). For fire safety, the team shall review the exits mentioned previously to make it clear to everyone where they are.

2.1.3 Preliminary Safety Data Sheets: [PLA+ SDS](#), [TPU SDS](#), [PETG SDS](#)

2.1.4 The team shall implement safety by including a section in the spreadsheet that lists all safety equipment necessary for each material, in accordance with the above-mentioned hazard analysis. The Safety Officer shall curate this and have the other team members check it to ensure it includes all necessary safety equipment, hazard mitigation strategies, and safety guidelines. Each team member must refer to this spreadsheet before using materials. After checking the spreadsheet, each team member shall add their name to the names column, acknowledging that they are using the material and have taken the necessary precautions listed on the sheet. In extreme cases, such as the rover catching on fire, the team shall use the fire extinguisher and, if needed, leave through the previously mentioned fire exits and pull the fire alarm. Additionally, the robot will have a physical kill switch in case it malfunctions.

2.2 In the first week, before any work begins at the DaVinci Workshop, the team will conduct a general safety briefing involving all personnel, during which we will list all hazards for each material in accordance with FMEA analysis and record them in the above-mentioned spreadsheet, along with hazard mitigation strategies and safety guidelines. Additionally, the Safety Officer shall reiterate key points of DaVinci Workshops' own safety plans.

When fabrication starts, the team shall hold a safety briefing on the equipment needed; any personnel who intend to work on the physical building of parts and the rover must attend. The team shall also have the advisor brief the team on how to use the tools, including the power tools, if necessary, and inform the team of any other points not addressed in the briefing.

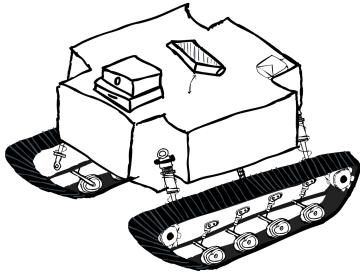
Similarly, when construction starts, the team shall have a similar-length briefing on how to use different tools and best practices for assembling the vehicle, with the same group of personnel present, and the advisor, if needed, shall discuss any safety-related points during construction.

General briefing framework: Before each new step of the building process, the team shall take 30 minutes to an hour to review the step, add any needed materials, and identify their hazards and required equipment.

2.3 For Personal Protective Equipment (PPE), we intend to include it in every necessary step, especially when we use power tools and during fabrication, which are the most dangerous steps for our project, as well as during design and construction. We will implement PPE through our spreadsheet system, in which every team member will sign off on the materials and tools they use and confirm they understand the required PPE. Additionally, for critical activities, including but not limited to fabrication and the use of power tools, the Safety Officer shall be present to ensure the necessary PPE is enforced. During driving and testing, the Safety Officer shall personally supervise PPE use when necessary, such as electrical gloves, rather than relying on the spreadsheet system. The same shall apply to the actual competition, where the Safety Officer shall brief the team on the required PPE and ensure everyone wears the proper equipment.

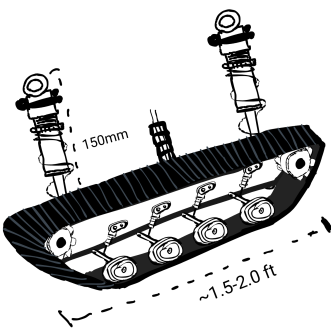
3. Technical Design

3.1 Our initial thoughts are that the rover will use a dual-suspended tank-tread system. Because treads provide superior traction and weight distribution, the rover should traverse obstacles with loose particles or debris more efficiently, such as obstacles 3, 6, 7, 8, and 9. The improved traction and weight distribution of the treads will also allow the rover to scale inclines better, such as in 4, 7, and 10, and the tracks will be wider than the crevasses in obstacle 5,



allowing it to cross with ease. The electronics will be mounted in the robot chassis, except for the Raspberry Pi camera and ToF sensor, which will have their own rotating housing. A mounted touchscreen will display results and possibly a camera feed during manual operation to help the driver aim the camera on the Y-axis before setting up for the task.

3.2 Using a design common for treads, we use four roller wheels, tensioned with ~48 mm shock absorbers, as the primary suspension system. This allows the treads to conform to the terrain. The rollers will be 3D printed in PETG, and the bars holding them will be cut from aluminum and attached to the commercially available shock absorbers. For the tracks, instead of a continuous loop of flexible material, we use a segmented track for durability. For early prototypes, we will print these in PETG or TPU and hold them together with metal dowels from the DaVinci Workshop. Later, we will likely replace these segments with lightweight aluminum, either by sending them to a third party for custom machining or creating them using tools at Olin College. We require the treads to be wider than 4/3 inch to easily pass the crevasses of Obstacle 5. Together, these parts are called tread assemblies, and we will use one per side, mounted on 150 mm shock absorbers to allow them to “rotate” forward and back and left to right, as explained next.



3.3 In addition to the primary suspension discussed in the last section, we also use a secondary suspension mechanism designed to handle vibrations and jitter that the first mechanism cannot catch, including wider vibrations caused by terrain. They also serve to widen the rover's base for stability. This mechanism puts the tread assemblies on either side of the rover on two spring suspension assemblies, as shown. Each tread will have two drive sprockets, each driven directly by a 12V 10:1 gearmotor ([Pololu Item 4748](#)). We will machine or laser-cut the drive sprockets from metal.

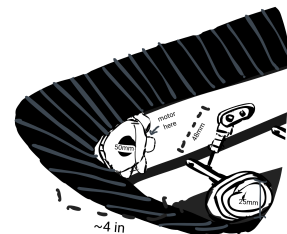
We estimated a no-load top speed of ~5.86 mph. We calculated this estimate

using the linear surface velocity formula, where D = diameter in meters, using a 50 mm sprocket.

$$V = \frac{RPM \cdot \pi D}{60} \Rightarrow V = \frac{1000 \text{ rpm} \cdot \pi \cdot 0.050 \text{ m}}{60} = 2.618 \text{ m/s}$$

$$2.618 \text{ m/s} \cdot 2.237 = 5.856 \text{ mph} \approx 5.86 \text{ mph}$$

Notice how this is above the 5 mph max. This speed will likely drop below 5mph with weight, but if not, we will use software limiting with GPS velocity calculations to stay close to, but not over, 5mph. If the GPS signal is lost, we will fall back to a single power value that guarantees high speed on flat ground.



3.4 The rover shall receive APRS commands using a 2m VHF receiver. To decode the data, the rover will use a software-defined TNC called Dire Wolf on the Raspberry Pi, which converts the radio's audio signal into the individual commands specified in the handbook for execution using the task completion mechanisms and software. Each NASA command will be saved and queued for execution only after the previous one has finished.

Our task completion mechanism centers on a gimbal-mounted camera on the chassis. After the starting AAA sequence, this gimbal will be able to independently rotate the X-axis to respond to NASA's FOV commands. We are also considering adding Y-axis rotation. Since the handbook says that the targets' shapes will be "at least one foot" above the ground, the exact height will be unknown. An independently rotating Y-axis will let the driver point the camera to the exact height before starting the task. Before the camera takes a photo, the rover will perform color filtering on command, either red or blue (overwriting the previous filter), depending on whether the rover receives E or F. The rover will use software filters instead of physical ones to save weight and complexity.

To measure the distance in feet to a target on command H, we will mount a ToF (Time of Flight) distance sensor on the camera gimbal. After taking a photo, we run a program to get the time, date, and distance (if applicable) and stamp them onto the photo. The rover will use a touchscreen to provide task results to judges during PER and show a live camera feed while driving. This feed will allow accurate Y-axis aiming before tasks begin. After all commands are finished, NASA will send a ZZZ ending sequence, and the rover will finish executing its task.

3.5 The largest technical challenge we will face is likely mitigating debris ingress into the treads. Any piece that gets into the right place or is large enough could seize the drive mechanism and lock up the tread assembly. The rover chassis will be angled at the front and rear to maximize usable clearance, with the suspension assemblies mounted behind the angled portions, as shown on the prior diagrams. This minimizes the risk of debris lodging between the springs and the chassis; however, this is only one mechanism we can use. To further reduce this risk, we could add a rigid side skirt to deflect inbound debris away from the treads.

In addition, our drive system naturally resists debris ingress. Since the sprocket is not solid, but two discs locked together, there are fewer places for debris to get stuck. The sprocket spokes should be able to puncture any debris caked on the treads. Also, since each roller has its own suspension, the wheels should roll over dirt or debris caked on the inside of the treads.

Additionally, we may encounter a problem with the ToF sensor. Since it must be aimed directly at the target, its beam may miss, making it less accurate than the camera. Thus, we may include software-derived distance as a backup or replacement. We will likely use this software-derived distance only if the ToF sensor's readings differ drastically from the math, indicating the ToF sensor was not aimed properly.

4. Project Plan

4.1 The following is a complete timeline for completing this project. Dedicated meeting times are not individually marked, but are planned for Mondays 4:00-6:00 PM and Tuesdays 5:00-6:00 PM (and possibly later). We also plan to meet at least once a weekend, though the exact time will likely vary week to week. We will continue to design as if we have been accepted into the program from proposal submission to selection.

	August	September	October	November	December
HERC	13 th : Handbook Released 20 th : Info Webinar	17 th : Proposals Due	8 th : Proposal Selection Announced 13 th : Kickoff Webinar & Team Socials Due	24 th : DRR & Slides Due 30 th : DR Presentations	
Design		6 th : Finish Proposal Draft 13 th : Finish Proposal 13 th -17 th : Proposal Buffer Days	9 th : Architecture Designs Due 16 th : Circuit Diagrams Due	3 rd : CAD/Code Draft Due 10 th : DR Slides Draft Due 12 th : CAD/Code Final Due 14 th : DRR Draft Due 19 th : DRS Final Due 21 st : DRR Final Due 25 th : Begin DRP Practice	 2 nd : Parts Deadline & Begin RC Code 4 th : Begin Final Construction
Building/Testing			17 th : Begin DRR 21 st : Begin DR Slides 25 th : Code Control Scheme Drafted	13 th : Order Parts	13 th : Begin Drivetrain & Task Execution 21 st : Drivetrain Due 23 rd : Task Execution Code Due 24 rd : Begin Task Execution Hardware & Misc Code
Engagement		18 th : Decide Social Media Strategy 20 th : Subsystem Assignments Final TBD: Weekly mentor meetings	11 th : Finalize Profiles 20 th : Finalize Sponsors TBD: Weekly mentor meetings Posts: 18 th , 25 th , 29 th	TBD: Weekly mentor meetings Posts: 1 st , 5 th , 8 th , 12 th , 19 th , 22 nd , 29 th	22 nd : Peter J. Eloranta Deadline TBD: Weekly mentor meetings Posts: 6 th , 13 th , 20 th , 24 th , 27 th

	January	February	March	April
HERC	21 st : ORR Expectations Webinar	9 th : Team Photos & Travel Rosters Due 22 nd : ORR, ORS, & Rover Photos Due	1 st -6 th : ORR Presentations	21 st -24 th : HERC Final
Design	22 nd : Begin ORR & ORS Drafts 6 th : Task Execution & Misc Code Due 12 th : Core Autonomy/Code	5 th : Begin Travel Roster & Team Photos 15 th : ORR/ORS Drafts Due 20 th : ORR/ORS/Rover Photos Final Due 23 rd : Begin ORP Practice		
Building/Testing	7 th : Begin Testing (Pilot & Autonomous) 26 th : Task System Tests 28 th : RC System Tests 29 th : Revised Code Due	1 st /8 th : Full-Rover Tests 10 th : Final Code Due	14 th : Competition-Ready Rover 15 th Onwards: Driver Practice	11 th : Final Testing & Pack For Travel 17 th : Rover Shipped To Huntsville
Engagement	TBD: Weekly mentor meetings & outreach event Posts: 3 rd , 7 th , 10 th , 14 th , 17 th , 20 th , 24 th , 28 th , 30 th	TBD: Weekly mentor meetings & outreach event Posts: 4 th , 7 th , 11 th , 14 th , 17 th , 21 st	TBD: Weekly mentor meetings Posts: 7 th , 12 th , 17 th , 21 st , 28 th	Posts: 4 th , 7 th , 14 th , 18 th , 19 th , 20 th

4.2

Bill of Materials

Item Name	Subsystem	Cost	Weight (lbs)	Notes
Aluminum U-Channel	Chassis	FREE (\$54)	0.0424/ inch	Available from our DaVinci Workshop/DVW (school makerspace). If limited supply/not available, \$0.45 per inch (120 inches = \$54)
Screws, nuts, square brackets, etc.	Chassis	FREE	~0.25	All are 100% available from DVW
1x Circuit breaker	Chassis	\$56.99	0.50	For safety issues and to prevent circuit overloads by cutting power Satisfies 6.4.7
PLA filament SDS	Chassis, Drivetrain	\$13.99 (Per kg)	~1-2	For prototyping the chassis and rigid parts of the track systems. Will likely be replaced by metal 3D printing from a service like PCBWay in the final rover
TPU filament/rubber SDS	Drivetrain	\$52.24 (8.81 lbs)	~1-2	For 3D printing the Rover's treads and prototyping 50mm sprockets.
5x DC Gear Motors	Drivetrain	\$179.15	1.93	4 motors (2 on each side, 1 front & back) for the rover and 1 extra. While it can go faster than 5 mph, software limiters will help satisfy 6.4.4
4x 50 mm sprocket	Drivetrain	\$99.96 (12 inches)	1.76	Prototyped with TPU. Later versions will be aluminum and custom-machined for prototyping at a local makerspace. (4pc. 3x3x3in)
2x Motor controllers	Drivetrain	\$46.96	0.5	Controls the speed and direction of the DC Gear motors.
Motor mounting brackets	Drivetrain	N/A	~1 (~0.25X2)	Custom brackets that mount the motors to the suspension frames while allowing the drive sprockets to rotate around them. These will be printed in PLA for prototyping and will likely be printed in metal using a service like PCBWay for the final rover.
2x Shock Absorber (150mm)	Drivetrain	\$79.36	0.42	Part of the rover's suspension system, responsible for providing suspension to the tracks as a whole
Shock Absorber (48mm)	Drivetrain	\$16.99 (Pack of 4)	0.13	Part of the rover's suspension system. Provides suspension for individual rollers to handle uneven terrain. May be too large, so may be replaced by 25mm Shock Absorbers (\$24.99, pack of 4)
1x Cable Chain (10x15mm 1.5 meters)	Drivetrain	FREE (\$19.49)	0.63	Part of the rover's suspension system, responsible for protecting cables from motors running back to the chassis. Most likely available at DVW
Ultimate GPS Breakout	Drivetrain	\$29.95	0.19	Allows exact ground speed calculations via satellite triangulation (GLONASS + GPS). From this, we can make a dynamic speed/power script to keep our rover close to 5 mph without going over.
External GPS Antenna	Drivetrain	\$21.50	0.44	Works with the Ultimate GPS Breakout to extend GPS gain by 28 decibels. Would make our speed/power script far more accurate and reliable by greatly improving the performance of the GPS/GLONASS signal
SMA to u.FL cable	Drivetrain	\$3.95	0.01	Intermediate cable between Ultimate GPS Breakout and External GPS Antenna

1x Battery Pack (11.1V 5000mAh)	Power	\$56.99 (FREE)	0.3	Powers the rover. Potentially some extras available at DVW. Satisfies 6.4.1
1x BMS (12V)	Power	\$11.99 (4 pack)	0.066	Battery Management System for protecting rechargeable batteries and cell balancing.
Insulated battery enclosure	Power	\$45.72	~025-0.5	Custom-designed enclosure to store the battery on the rover. Printed in PETG at DVW, with EVA foam & Reflectix on the inside for insulation. Satisfies 6.4.6
Connector plugs/wires	Power	FREE	~1	All available at DVW (male & female)
1x Voltage Regulator	Power	\$8.99	0.024	5V 3A External UBEC: Regulates high input voltages
RC transmitter	Task	\$54.89	N/A	Controller for driving the rover around obstacles
1x RTC module	Task	\$14.95	0.0062	Keeps time automatically so the date/time on task photos is correct
2m VHF receiver	Task	\$14.65	0.022	For receiving FM signals in the 144-148 MHz band.
Raspberry Pi 4 (4GB)	Task	FREE (\$35.00)	0.1	Mini computer for decoding task: Already have
Pi Camera Module 3	Task	FREE (\$34.99)	0.0088	Camera for taking images of the assigned targets: Already have
TOF Sensor	Task	\$23.34	0.008	Used to detect distance to target
Pi Touch Display 2	Task	\$60.00	0.5577	Used to display live camera feed to allow the pilot the ability to aim up/down and to display images taken to judges in PER
Fasteners, wires, etc.	Misc	FREE	~1	All are available in DVW.
Assorted tools/materials (caliper, PETG, pliers, etc.)	Misc	FREE	N/A	Available in DVW (main option) and a local makerspace (contingency plan) SDS
Alternate Makerspace	Misc	\$280 (7 days)	N/A	Workspace with tools & equipment to build the rover. Day pass is \$40. Located in Brighton.
PCBWay	Misc	~\$50 - \$200	~0.5-2	For producing custom mounting brackets and other complex metal parts in later variations of the rover chassis.
Flights	Travel	~\$2800	N/A	We estimate 7 people will go to Huntsville, AL. Flights seem to be about \$400 per person as of 8/22/2026, which comes to ~\$2800.
Hotels	Travel	\$1727 (\$265 a night)	N/A	With 7 people, we will need 2-3 rooms. We will be staying 5 nights from 4/20/26-4/25/26.
Additional expenses/mass buffer	Travel/ Misc	~\$945	~5	Includes estimated food, transportation, etc. (\$45 x 3 days x 7 people)

Totals

*(Low end cost means reusing existing materials, high end cost means buying everything new) **(Low end mass is taken from the bottom of ranges, high end mass is taken from the top of ranges)				
Subsystem	Low End Cost*	Low End Mass**	High End Cost*	High End Mass**
Chassis	\$70.98	6.838 lbs	\$124.98	7.838 lbs
Drivetrain	\$530.06	8.01 lbs	\$557.55	9.01 lbs
Power	\$66.70	1.64 lbs	\$123.69	1.89 lbs
Task	\$167.83	0.702 lbs	\$237.82	0.7027 lbs
Miscellaneous	\$50.00	6.5 lbs	\$480.00	8.0 lbs
	Low End Total w/ est. tax		High End Total w/ est. tax	
Base & contingency (+15%)	\$940.92 (\$1,082.06)		\$1619.29 (\$1,862.19)	
w/ Travel (~\$5,472 before tax)	\$6,939.92		\$7,091.29	
	Low End Mass Total		High End Mass Total	
Base & w/ Buffer	18.691 lbs (23.691 lbs)		22.4407 lbs (27.4407 lbs)	

4.3 Confirmed amounts: We currently have a pledge of \$2000 from Arrow Intelligent Systems Inc. They have also pledged to match other donors' contributions for up to \$1000.

Fundraising plans: We will reach out to companies likely to be interested in this effort. Specifically, we plan to contact tech companies in our local area and B2B software companies (AutoCAD); Massachusetts Science & Engineering Fair sponsors; tool companies (i.e., DeWalt, Home Depot); Car companies (i.e., Mercedes, Honda, McLaren, Toyota); Racing companies (i.e., Red Bull and Mercedes); Aerospace (Boeing, SpaceX, etc.); existing sponsors of our High School's robotics teams (REV Robotics, PTC Group, etc.); and local companies like MathWorks. Because we are located in a high-tech corridor (Route 128 Corridor), we have numerous fundraising opportunities with these companies, increasing our chances of securing a donation.

Local area banks have certain amounts earmarked, typically for community investments. We plan to reach out to the community investment arm of local banks (i.e., Needham Bank and Rockland Bank). We also plan to contact Olin College of Engineering near us for material sponsorship, including access to manufacturing facilities. We expect to fund our budget through these efforts. We created a dedicated website to attract sponsors and expand our team outreach.

4.4 Needham High School's Student Council provides funding to clubs that apply. Due to the team's size, we expect to receive ~\$250 from them. Additionally, the Superintendent supports us and may connect us with potential funding partners. Our team is also eligible for our school's Peter J. Eloranta Scholarship, which provides partial or full funding for educational opportunities outside regular classes. We plan to apply when it opens in early December and expect it will cover ~\$3,000 of the total costs. Our school does not fund clubs and teams through basic donations.

5. Engagement; STEM Industry Engagement Plan

5.1 Within our community, we have several contacts and potential partners. For example, we contacted **Rohit Viswanathan**, a highly experienced Northrop Grumman Consulting Engineer who contributed to operational research studies at FAA and NASA and led the Integrated Product Team on HALO, part of the Gateway lunar mission of Artemis. He has agreed to help mentor and advise us on various aspects of the challenge (see 5.2). We also have connections with **Duong Vu**, a senior software engineer; **Bao Nguyen**, Principal Application Engineer; and **Ryan Chladny**, a robotics club mentor and Engineering/Software Development Manager, all of whom work at MathWorks. They can teach us not only software engineering but also the basics of MATLAB and Simulink, which are essential for modern engineering. Because some of our members previously won the MIT BWSI Build a CubeSat Challenge, we also have a connection to **Urban Gillespie** at Lincoln Labs, an aerospace engineer and former Director of Command & Control Development Programs at the Missile Defense Agency. He has already taught us a lot about aerospace design and will continue to do so. We will also collaborate with **Greg Jerman**, a Failure Analysis Subject Matter Expert at NASA. He has over 30 years of experience in various propulsion hardware technologies, analytical microscopy, and ISS hardware. He can teach us how to learn from failures in our rover. Since ML is important for system control, potential partnerships through contacts at **Arrow Intelligent Systems** may help us learn more. Additionally, we may contact professors at our local engineering school, **Olin College**, for expert advice on a problem.

5.2b As described in 5.1, we have various local contacts we can reach out to. MathWorks would provide experience with MATLAB, which many STEM companies use, and simulation. We will aim to get certification in MATLAB and Simulink. Furthermore, we have connections with Urban Gillespie. Mr. Gillespie will continue to offer guidance and teach technical concepts relating to aerospace. His practical experience in engineering environments also allows him to help with fundraising and presentations. One of our members has personal connections with Rohit Viswanathan from Northrop Grumman, who will offer valuable insight into practical aerospace engineering. We can learn critical failure-analysis techniques from Greg Jerman at NASA. Our connections at Arrow Intelligent Systems would provide machine learning experience. Additionally, we are working to build connections with Olin College, whose mentors would provide experience, technology, and connections for future opportunities.

5.3b Our team plans to spread awareness locally and on social media. Although we have a dedicated [website](#), our primary platform will be [Instagram](#), and we plan to post regularly (see 4.1). We will target youth in our community to encourage younger audiences to pursue robotics. We also plan to visit parts of our community, such as elementary schools and our public library, to do outreach, such as demonstrating our prototype and/or finished rover.



NEEDHAM HIGH SCHOOL

A CARING COMMUNITY
STRIVING FOR PERSONAL GROWTH AND
ACADEMIC EXCELLENCE

September 1, 2026

To Whom It May Concern,

On behalf of Needham High School I am pleased to express my support for the following Needham High students Hai Dang, Itay Kedem, Karan Krishnan, Christos Kyttaris, Sophie Liu, Alex Shaigorodsky, and Zirui Wang and their participation in the NASA Human Exploration Rover Challenge (HERC).

The team is composed of students with demonstrated experience and interest in engineering, robotics, aerospace, and STEM education. Several members of the team, including Zirui, Christos and Karan previously participated in the MIT Beaver Works Summer Institute (BWSI) CubeSat program and were members of the team that won the national competition. Their prior experience reflects both their technical capabilities and their commitment to collaborative, hands-on engineering.

Through HERC, the students will design, prototype, build, and test a remotely operated lunar rover capable of navigating obstacles representative of extraterrestrial terrain. The team will also address an autonomous image-identification component of the challenge. Participation in this competition will provide students with valuable opportunities to apply concepts from engineering, computer science, robotics, physics, and mathematics to a challenging real-world problem.

We understand that the competition will involve multiple stages through April, beginning with the proposal submission in September and culminating, for qualifying teams, in a final competition at the U.S. Space & Rocket Center in Huntsville, Alabama. We are supportive of the students' efforts to participate throughout this process and recognize the significant educational value of undertaking a project of this scope.

The students plan to conduct the majority of their design, construction, and testing activities through the DaVinci Workshop, with potential access to additional manufacturing resources through Olin College of Engineering as needed. Mr. Hans Batra will serve as the team's advisor throughout the competition.

Needham High School supports the team's application to the NASA Human Exploration Rover Challenge and is willing to provide reasonable institutional support to enable the students to participate in the competition. We understand that the students are independently pursuing fundraising to cover project materials and potential travel expenses associated with the final competition.

We are proud to support our students as they pursue this opportunity and believe that participation in HERC will encourage technical innovation, teamwork, problem-solving, and continued interest in STEM fields.

Sincerely,

Aaron Sicotte, Principal